PAST-Bench: Benchmarking the Foundations of Recursive Self-Improvement in Personal Agents

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HarnessRSI 71 ★★★

evidence: retained

Paper

Code

1 What is tested; Hermes+

26 task families, 204 episodes, fresh session each

Cold → Learn → Eval → Control

context wiped between episodes; Eval omits the trigger wording

w/ evolve	w/o evolve
may read memory, skills, sessions	same prompt, grader, tools, seed

Δ = S(w/ evolve) − S(w/o evolve), plus Mech trace score

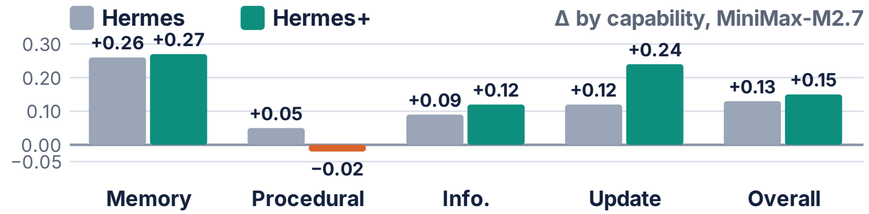
Hermes+ = Hermes + 5 loop mechanisms

- **E1 Plan** — check saved memory and skills before drafting a recall-dependent action
- **E2 Render** — typed memory (scope, value, supersedes, expiry); show only the current binding
- **E3 Route** — save procedures as ranked, patchable skills
- **E4 Gate** — withhold a draft answer once and prompt a session or skill lookup if none has happened
- **E5 Close** — at episode end write the corrected binding, overwrite the old one, flush

no verifier described (our reading): the agent decides what to save
the benchmark measures whether keeping helped

2 Headline: persistence helps, unevenly

Hermes, 7 models: Overall Δ **+0.13 to +0.24**. Memory +0.23 to +0.37, Update +0.12 to +0.36, Procedural +0.03 to +0.19, Info +0.05 to +0.11.



Overall Δ 0.13 ± 0.04 vs 0.15 ± 0.06 (3 runs): not read as a stable gain. Same Δ, different Mech: nanobot +0.13 / 0.57, Hermes +0.13 / 0.64. Hermes+ +0.15 / 0.73; Agent-Zero −0.08 / 0.39.

3 Controls that decide if the gain is real

Single-mechanism ablation · MiniMax-M2.7

Table 4 (one mechanism added)	Overall w/	Overall Δ
Base Hermes	0.66	+0.13
+ E1 Plan	**0.62**	+0.14
+ E2 Render	**0.60**	+0.13
+ E3 Route	**0.51**	+0.12
+ E4 Gate	**0.55**	+0.17
+ E5 Close	**0.55**	+0.12
Hermes+ (all five)	0.66	+0.15

Each mechanism helps its target (E2 Memory w/ 0.80, E3 Procedural Δ +0.10, E4 Info Δ +0.17, E5 Update Δ +0.16) but lowers Overall w/ (red); E4 alone Δ +0.17 > all five.

Hermes+ on other base models (Table 6)

Overall Δ, other models	Hermes	Hermes+
MiniMax-M2.7 (tuned on)	+0.13	**+0.15**
DeepSeek-V4-Pro	+0.17	**+0.15**
GPT-5.4	+0.24	+0.24
Claude Sonnet 4.6	+0.20	+0.22
Claude Opus 4.6	+0.19	**+0.18**

3 of 5 matched or improved. Mech: 0.64 → 0.73 on MiniMax; 0.71 → 0.69, 0.80 → 0.80, 0.76 → 0.77, 0.70 → 0.70 on the rest (table order). Mech rise may be partly by construction: E1/E4 force reads, E2/E3/E5 writes (our reading). Mechanisms not given to other frameworks.

4 Where the gain is absent or thin

- **Procedural:** Hermes+ persistence-on 0.55 → 0.38 (−0.17‡), Δ +0.05 → −0.02
- **Update gain:** dominated by fact correction (−0.078 → +0.492); 5 of 7 families rose, 2 fell (recall-then-modify +0.347 → +0.190, SOP-patch 02 +0.017 → −0.127) in 7–8-episode families. Other 4 models: Update Δ changes by −0.05‡ to +0.05‡
- **Cost (MiniMax-M2.7):** 31,859 vs 12,615 tokens/episode (≈2.5×; other models 0.55‡×–1.84‡×), no matched-token row; Agent-Zero 1,200 s vs 300 s
- Synthetic tasks (made with GPT-5.4 / Opus 4.6 agents); judge = tuning model; no control scores or held-out split

Takeaway

Retained experience helps every tested model, mostly on Memory and Update, and equal Δ can hide different pathway evidence. Hermes+ lifts Update on its tuning model; its overall gain (+0.15 vs +0.13) is inside run-to-run variation, at about 2.5× the tokens.

Our read · 71 ★★★ · retained

Reports: matched on/off, 3-run SD (2 configs), ablations, cost, code.
Does not report: held-out split for Hermes+, control scores, matched-compute row.